CE2005
Linear Algebra

Linear Independence and Bases

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Intended Learning Outcomes

By the end of this lecture, you should be able to:

- Explain linear dependence and independence, and relate them to geometry and minimal spanning sets.
- Test vector independence with homogeneous systems and determinants for square matrices, and explain when a representation as a linear combination is unique.
- Test independence in polynomial and function spaces, and explain what a Wronskian can establish.
- Verify a basis, determine dimension, and justify extending independent sets or reducing spanning sets to a basis.

Outline

1. What Do Linear Dependence and Independence Mean?
2. How Do Linear Systems Test Independence and Explain Unique Representations?
3. How Do We Test Independence of Polynomials and Functions?
4. How Do Bases and Dimension Describe a Vector Space?

What Do Linear Dependence and Independence Mean?

From Spanning Sets to Linear Independence

- In this section, we look more closely at the structure of vector spaces. We consider only finite generating sets.
- A generating set is a spanning set.
- Each vector in the vector space can be built up from its elements using addition and scalar multiplication.
- A minimal spanning set has no unnecessary elements: all its elements are needed to span the vector space.
- We must therefore examine how its vectors depend on each other. This connects spanning sets from Section 3.2 to linear dependence and independence.

A Spanning Set with an Unnecessary Vector

- Consider the following vectors in R^3:

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} -1 \\ 3 \\ 8 \end{bmatrix}$$

- Let S be the subspace of R^3 spanned by x_1, x_2, and x_3.
- The third vector is already in the span of the first two:

$$x_3 = 3x_1 + 2x_2.$$

- Indeed, its entries are $3 - 4 = -1$, $-3 + 6 = 3$, and $6 + 2 = 8$.
- Thus, x_3 adds no new vectors to S.

Reducing a Linear Combination

- Any linear combination of x_1, x_2, and x_3 reduces to a linear combination of x_1 and x_2:

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3$$

$$= \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 (3x_1 + 2x_2)$$

$$= (\alpha_1 + 3\alpha_3) x_1 + (\alpha_2 + 2\alpha_3) x_2.$$

- Thus, $S = \text{Span}(x_1, x_2, x_3) = \text{Span}(x_1, x_2)$.
- Every combination of the three vectors is a combination of the first two.

  Conversely, a combination of x_1 and x_2 is a combination of all three with coefficient 0 on x_3.

The Dependency Relation

- The relation $x_3 = 3x_1 + 2x_2$ can be written as

  $3x_1 + 2x_2 - x_3 = 0.$

- All three coefficients in this relation are nonzero, so

  $x_1 = -(2/3)x_2 + (1/3)x_3,$

  $x_2 = -(3/2)x_1 + (1/2)x_3,$

  $x_3 = 3x_1 + 2x_2.$

  $\text{Span}(x_1, x_2, x_3) = \text{Span}(x_1, x_2) = \text{Span}(x_1, x_3) = \text{Span}(x_2, x_3).$

  Any two of these three vectors span the same subspace S.

No Dependency Relation for the First Two Vectors

- Suppose $c_1x_1 + c_2x_2 = 0$, where c_1 and c_2 are not both 0.

  If $c_1 \neq 0$, then $x_1 = -(c_2/c_1)x_2$.

  If $c_2 \neq 0$, then $x_2 = -(c_1/c_2)x_1$.
- Either possibility makes one vector a multiple of the other.
- But the entries of x_1 and x_2 show this is impossible:

  the first entries would require the multiple $-1/2$,

  whereas the second entries would require $-1/3$.
- Therefore, $c_1x_1 + c_2x_2 = 0$ implies $c_1 = c_2 = 0$.
- Span(x_1) and Span(x_2) are both proper subspaces of Span(x_1, x_2); both vectors are needed.

Two Observations about Spanning Sets

(I) If $v_1, v_2, \ldots, v_n$ span a vector space V and one of these vectors can be written as a linear combination of the other $n - 1$ vectors, then those $n - 1$ vectors span V.

(II) Given n vectors $v_1, \ldots, v_n$, it is possible to write one of the vectors as a linear combination of the other $n - 1$ vectors if and only if there exist scalars $c_1, \ldots, c_n$, not all zero, such that $c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0$.

The first observation concerns the span.
The second expresses a dependency using a zero linear combination.

Proof of Observation (I)

Suppose v_n is a linear combination of the other vectors:

$$v_n = \beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_{n-1} v_{n-1}.$$

Let v be any element of V. Because $v_1, \ldots, v_n$ span V,

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_{n-1} v_{n-1} + \alpha_n v_n.$$

Substitute the expression for v_n into this combination:

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_{n-1} v_{n-1} + \alpha_n(\beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_{n-1} v_{n-1}).$$

Distributing α_n and collecting coefficients removes v_n from the representation without changing the vector v.

Completing the Proof of Observation (I)

Collecting the coefficients from the substitution gives

$$v = (\alpha_1 + \alpha_n\beta_1)v_1 + (\alpha_2 + \alpha_n\beta_2)v_2 + \cdots + (\alpha_{n-1} + \alpha_n\beta_{n-1})v_{n-1}.$$

This is a linear combination of $v_1, \ldots, v_{n-1}$.

The vector v was an arbitrary element of V, so every vector in V can be written using these $n - 1$ vectors.

Hence, $v_1, \ldots, v_{n-1}$ span V.

Deleting a vector already in the span of the others does not change the vector space that they span.

Proof of Observation (II)

First suppose one vector, say v_n, is a linear combination of the others:

$$v_n = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_{n-1} v_{n-1}.$$

Subtract v_n from both sides: $\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_{n-1} v_{n-1} - v_n = 0$.

Set $c_i = \alpha_i$ for $i = 1, \ldots, n-1$ and $c_n = -1$.

Then $c_1 v_1 + c_2 v_2 + \cdots + c_n v_n = 0$.

These coefficients are not all zero because $c_n = -1$.

Thus, a vector expressed using the others produces a nonzero choice of coefficients giving the zero vector.

The Converse in Observation (II)

Conversely, suppose $c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0$ and at least one of the coefficients, say c_n, is nonzero.

Move the other terms to the right-hand side:

$$c_nv_n = -c_1v_1 - c_2v_2 - \cdots - c_{n-1}v_{n-1}.$$

Because $c_n \neq 0$, divide by c_n:

$$v_n = (-c_1/c_n)v_1 + (-c_2/c_n)v_2 + \cdots + (-c_{n-1}/c_n)v_{n-1}.$$

This writes v_n as a linear combination of the others.

Both directions are established.

The condition “not all zero” is what permits this division.

Linear Independence

- Definition:

  The vectors $v_1, v_2, \ldots, v_n$ in a vector space V are said to be linearly independent if $c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0$ implies that all the scalars $c_1, \ldots, c_n$ must equal 0.

- Observation (II) shows that none of these vectors can be written as a linear combination of the others.
- To verify independence, we must consider every possible choice of coefficients giving the zero vector and show that all those coefficients must be zero.

Linear Independence and Minimal Spanning Sets

- If $\{v_1, v_2, \ldots, v_n\}$ is a minimal spanning set, then $v_1, v_2, \ldots, v_n$ are linearly independent. Otherwise, Observation (II) would express one vector using the others, and Observation (I) would let us delete it while still spanning V, contradicting minimality.
- Conversely, if $v_1, \ldots, v_n$ are linearly independent and span V, then $\{v_1, \ldots, v_n\}$ is a minimal spanning set. If one vector could be removed without changing the span, it would be a combination of the others, contradicting Observation (II) and independence.
- A minimal spanning set is called a basis.

Example 1

- Are the vectors $\begin{bmatrix}1\\1\end{bmatrix}$ and $\begin{bmatrix}1\\2\end{bmatrix}$ linearly independent or linearly dependent?

$$c_1\begin{bmatrix}1\\1\end{bmatrix}+c_2\begin{bmatrix}1\\2\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$$

Equating corresponding entries gives

$c_1 + c_2 = 0$

$c_1 + 2c_2 = 0.$

Subtracting the first equation from the second gives $c_2 = 0$;

substituting into the first gives $c_1 = 0$.

The only solution is $c_1 = c_2 = 0$, as required.

Linear Dependence

- Definition:

  The vectors $v_1, v_2, \ldots, v_n$ in a vector space V are said to be linearly dependent if there exist scalars $c_1, c_2, \ldots, c_n$, not all zero, such that $c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0$.
- Unlike independence, dependence is established by finding one choice of coefficients that are not all zero. Observation (II) then shows that at least one vector can be written as a linear combination of the others.
- The next example gives such a choice explicitly.

Example 2

- Are the vectors $\mathbf{x} = (1, 2, 3)^T$, $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ and $\mathbf{x}$ are linearly independent or linearly dependent?

$$c_1\begin{bmatrix}1\\2\\3\end{bmatrix} + c_2\begin{bmatrix}1\\0\\0\end{bmatrix} + c_3\begin{bmatrix}0\\1\\0\end{bmatrix} + c_4\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix} \Rightarrow \begin{bmatrix}c_1 + c_2\\2c_1 + c_3\\3c_1 + c_4\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

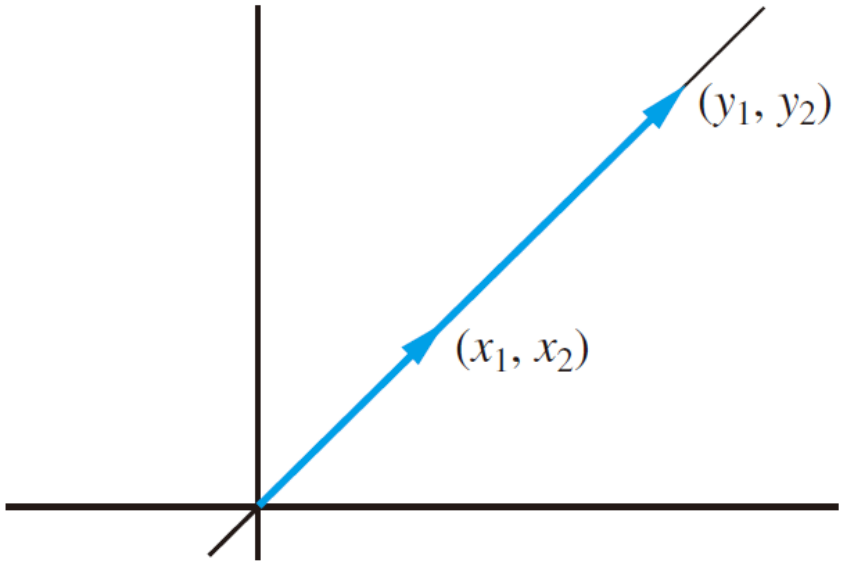
$$\Rightarrow c_1 = 1, c_2 = 2, c_3 = 3, c_4 = -1$$

$$\therefore \text{ linearly dependent}$$

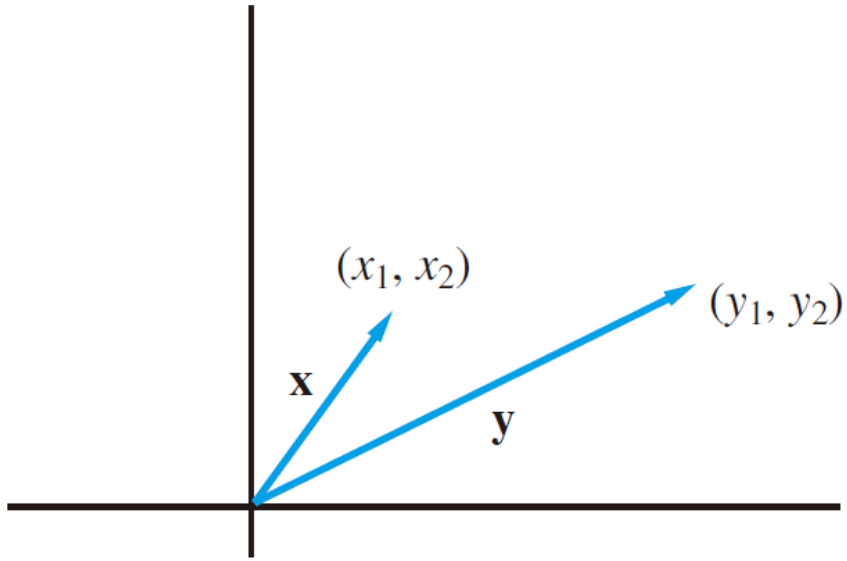
Trivial and Nontrivial Choices of Scalars

- For any set $\{v_1, v_2, \ldots, v_n\}$ in a vector space V, $c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0$ always holds when $c_1 = c_2 = \cdots = c_n = 0$. This is the trivial choice of scalars.
- If there are nontrivial choices of scalars for which $c_1v_1 + \cdots + c_nv_n$ equals the zero vector, then $v_1, \ldots, v_n$ are linearly dependent.
- If the only way this combination can equal the zero vector is for all the scalars $c_1, \ldots, c_n$ to be 0, then $v_1, \ldots, v_n$ are linearly independent.
- "Nontrivial" means at least one coefficient is nonzero.

Geometric Interpretation in R^2



(a) **x** and **y** linearly dependent



(b) **x** and **y** linearly independent

Figure 3.3.1.

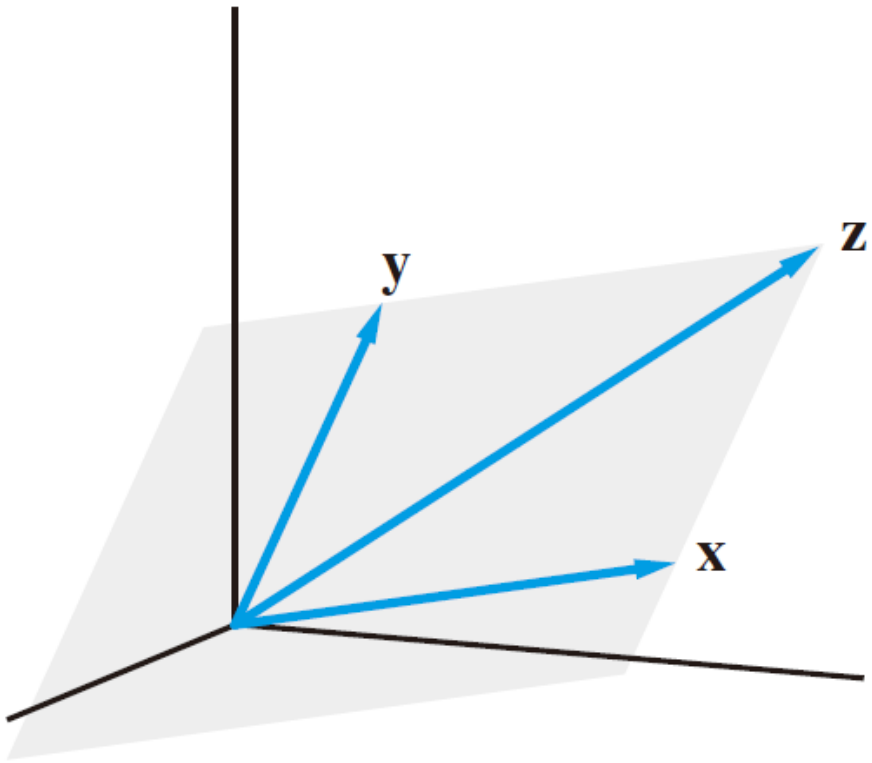
$c_1 x + c_2 y = 0$
c_1 and c_2 are not both 0,
say $c_1 \neq 0$, then
$x = (-c_2/c_1)y = k\,y$

$x \neq k\,y$

Two Independent Vectors Determine a Plane

- Let x and y be linearly independent vectors in $\mathbb{R}^3$: $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_2 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_2 \end{bmatrix}$
- The points (x_1, x_2, x_3) and (y_1, y_2, y_3) do not lie on the same line through the origin in 3-space.
- Because $(0, 0, 0)$, (x_1, x_2, x_3), and (y_1, y_2, y_3) are not collinear, they determine a plane.
- This plane is Span(x, y). We next consider where the point (z_1, z_2, z_3) lies relative to this plane.

A Third Vector in the Plane

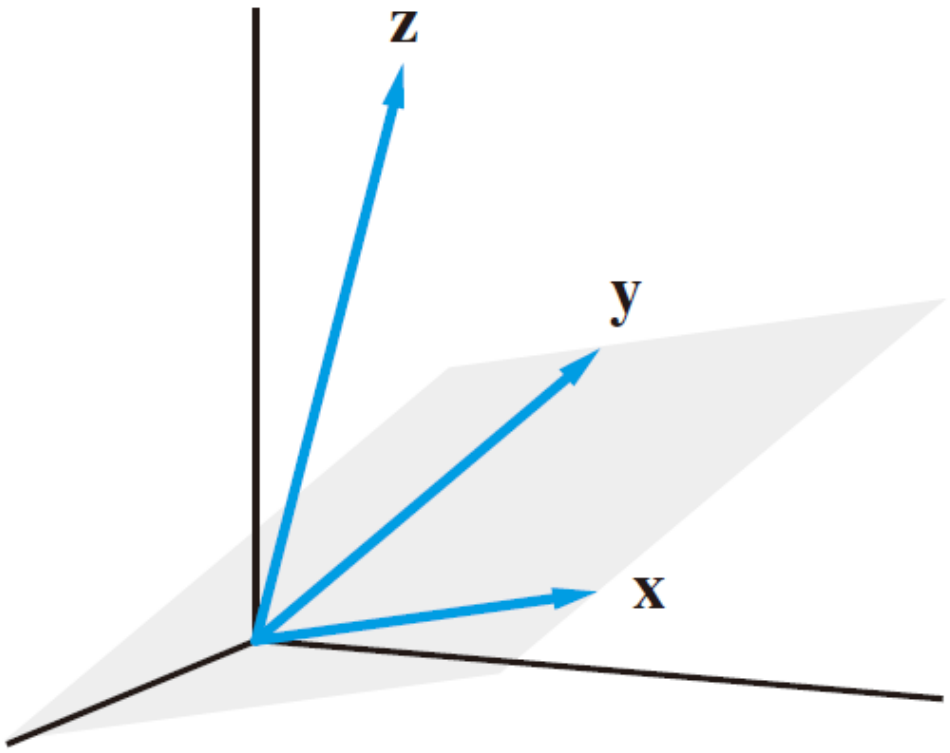


(a)

Figure 3.3.2.

- If (z_1, z_2, z_3) lies on this plane, then $z = (z_1, z_2, z_3)^T$ is a linear combination of x and y.
- Hence x, y, and z are linearly dependent.
- By Observation (I), adding z does not enlarge the span of x and y.

A Third Vector outside the Plane



(b)

- If (z_1, z_2, z_3) does not lie on the plane, then x, y, and z are linearly independent.
- Any nontrivial zero combination must involve z, because x and y are independent. Solving that relation for z would place it in Span(x, y), contrary to its position outside the plane.

How Do Linear Systems Test Independence and Explain Unique Representations?

Example 3 Independence in $\mathbb{R}^3$

Which of the following collections of vectors are linearly independent in $\mathbb{R}^3$?

(a) $(1, 1, 1)^T$, $(1, 1, 0)^T$, $(1, 0, 0)^T$

(b) $(1, 0, 1)^T$, $(0, 1, 0)^T$

(c) $(1, 2, 4)^T$, $(2, 1, 3)^T$, $(4, -1, 1)^T$

- For each collection, form a linear combination equal to zero.
- We must determine whether its coefficients must all be zero.
- The number of vectors need not equal the number of entries.

Example 3(a) Only the Trivial Solution

We must show that

$$c_1(1, 1, 1)^T + c_2(1, 1, 0)^T + c_3(1, 0, 0)^T = (0, 0, 0)^T$$

is possible only if the scalars c_1, c_2, c_3 are all zero.

Equating the three entries gives

$$c_1 + c_2 + c_3 = 0$$

$$c_1 + c_2 = 0$$

$$c_1 = 0$$

The third equation gives $c_1 = 0$; the second then gives $c_2 = 0$; the first then gives $c_3 = 0$.

The only solution is $c_1 = c_2 = c_3 = 0$.

Therefore, the three vectors are linearly independent.

Example 3(b) Two Independent Vectors

If

$$c_1(1, 0, 1)^T + c_2(0, 1, 0)^T = (0, 0, 0)^T,$$

then

$$(c_1, c_2, c_1)^T = (0, 0, 0)^T.$$

Equality of the entries gives $c_1 = 0$ and $c_2 = 0$.

The third entry repeats the condition $c_1 = 0$.

Therefore, the two vectors are linearly independent.

There are three entry equations but only two coefficients.

Independence requires that every coefficient be zero.

Example 3(c) A Singular Coefficient Matrix

If

$$c_1(1, 2, 4)^T + c_2(2, 1, 3)^T + c_3(4, -1, 1)^T = (0, 0, 0)^T,$$

then equating entries gives

$$c_1 + 2c_2 + 4c_3 = 0$$

$$2c_1 + c_2 - c_3 = 0$$

$$4c_1 + 3c_2 + c_3 = 0$$

The coefficient matrix is singular, so the homogeneous system has nontrivial solutions.

Therefore, the vectors are linearly dependent.

We next verify this conclusion by elimination.

Example 3(c) Verifying the Nontrivial Solutions

- Subtract twice the first equation from the second, and four times the first from the third. Divide the new second equation by −3, then add five times it to the new third equation.

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 2 & 1 & -1 & 0 \\ 4 & 3 & 1 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

- Thus, $c_2 = -3c_3$ and $c_1 = 2c_3$. The coefficient c_3 is free.
- Choosing $c_3 = 1$ gives $c_1 = 2$ and $c_2 = -3$, so
- $2(1, 2, 4)^T - 3(2, 1, 3)^T + (4, -1, 1)^T = (0, 0, 0)^T$.
- This is a linear combination with coefficients not all zero.

Theorem 3.3.1 Dependence and Singularity

Let $x_1, x_2, \ldots, x_n$ be n vectors in $\mathbb{R}^n$ and let $X = (x_1, \ldots, x_n)$. The vectors $x_1, x_2, \ldots, x_n$ will be linearly dependent if and only if X is singular (i.e., $\det(X) = 0$).

- Here X is the $n \times n$ matrix whose columns are the vectors.
- Example 3(a) has a nonsingular coefficient matrix;
- Example 3(c) has a singular coefficient matrix.
- The theorem explains both conclusions through the same test.
- Its hypothesis is n vectors in $\mathbb{R}^n$, so X must be square.

Theorem 3.3.1 Proof

The equation $c_1x_1 + c_2x_2 + \cdots + c_nx_n = 0$ can be rewritten as a matrix equation:

$$c_1\begin{bmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{bmatrix} + c_2\begin{bmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{bmatrix} + \cdots + c_n\begin{bmatrix} x_{1n} \\ x_{2n} \\ \vdots \\ x_{nn} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\Rightarrow c_1x_{11} + c_2x_{12} + \ldots + c_nx_{1n} = 0$$
$$c_1x_{21} + c_2x_{22} + \ldots + c_nx_{2n} = 0$$
$$\vdots$$
$$c_1x_{n1} + c_2x_{n2} + \ldots + c_nx_{nn} = 0$$

Let $\mathbf{c} = (c_1, c_2, \ldots, c_n)^T$, then the system can be written as $X\mathbf{c} = \mathbf{0}$

$\mathbf{c}$ has a nontrivial solution iff $\boldsymbol{X}$ is singular

$\Rightarrow \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ will be linearly dependent iff X is singular.

Testing n Vectors in $\mathbb{R}^n$ Using a Determinant

- Form a matrix X whose columns are the vectors being tested.
- Calculate det(X) to determine whether X is singular.

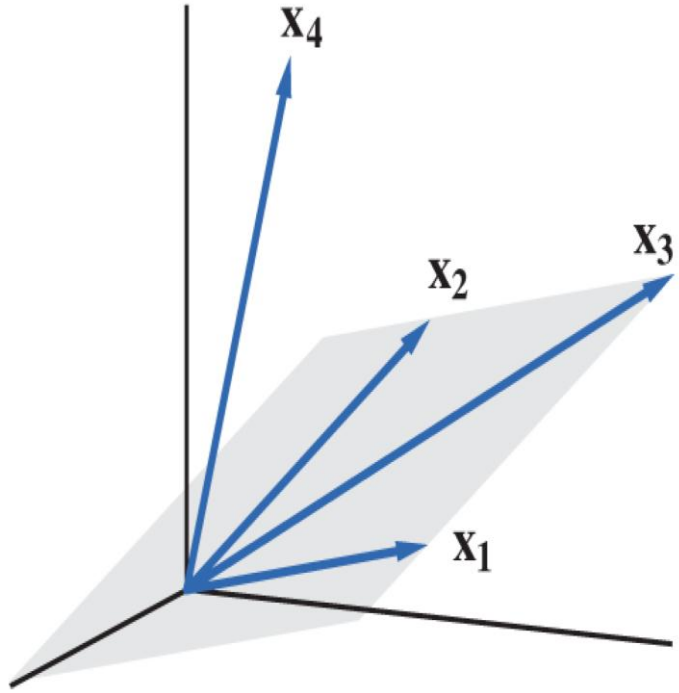
  If det(X) = 0, X is singular and the vectors are linearly dependent.

  If det(X) ≠ 0, X is nonsingular and the vectors are linearly independent.
- Theorem 3.3.1 connects this determinant calculation to the existence of nontrivial solutions of Xc = 0.
- The determinant test applies to the square matrix formed from n vectors in $\mathbb{R}^n$.

Example 4 Four Vectors in $\mathbb{R}^3$

$x_1 = (2, 3, 1)^T$, $x_2 = (3, 2, 1)^T$, $x_3 = (5, 5, 2)^T$, $x_4 = (2, 2, 4)^T$.

- The first three lie in one plane. The displayed relation is $x_3 = x_1 + x_2$.
- Since $x_3 = x_1 + x_2$, the vector x_3 lies in the plane spanned by x_1 and x_2.
- It follows that $x_1 + x_2 - x_3 + 0x_4 = 0$.
- The coefficients $c_1 = 1$, $c_2 = 1$, $c_3 = -1$, $c_4 = 0$ are not all zero, so the four vectors are linearly dependent.
- A dependency relation may have some zero coefficients.
- We do not need to use x_4 to establish this dependence.



Example 5 Applying the Determinant Test

Determine whether $(4, 2, 3)^T$, $(2, 3, 1)^T$, and $(2, -5, 3)^T$ are linearly dependent.

$$\begin{vmatrix} 4 & 2 & 2 \\ 2 & 3 & -5 \\ 3 & 1 & 3 \end{vmatrix} = 0$$

- Expanding the determinant along the first row gives

  $\det(X) = 4(9 + 5) - 2(6 + 15) + 2(2 - 9) = 56 - 42 - 14 = 0.$

- By Theorem 3.3.1, the vectors are linearly dependent.
- An explicit relation is $2v_1 - 3v_2 - v_3 = 0$.

Testing k Vectors in $\mathbb{R}^n$

- To test vectors $x_1, x_2, \ldots, x_k$ in $\mathbb{R}^n$, rewrite $c_1x_1 + c_2x_2 + \cdots + c_kx_k = 0$ as $Xc = 0$, where $X = (x_1, x_2, \ldots, x_k)$.
- If $k \neq n$, then X is not square, so determinants cannot be used to decide whether the vectors are independent.
- The homogeneous system always has the trivial solution $c = 0$.
- Its row echelon form has free variables if and only if there are nontrivial solutions; the vectors are then dependent.
- If there are no free variables, $c = 0$ is the only solution, and the vectors are linearly independent.
- Thus, elimination provides a test even when X is rectangular.

Example 6 Three Vectors in $\mathbb{R}^4$

Given the following vectors, determine whether they are linearly independent:

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} -2 \\ 3 \\ 1 \\ -2 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \\ 7 \\ 7 \end{bmatrix}$$

- There are three vectors, each with four entries.
- To test independence, solve $Xc = 0$, with these vectors as the columns of X.

  There are four equations in the three unknown coefficients c_1, c_2, c_3.
- Since X is 4×3, use elimination rather than a determinant.

Example 6 Reducing Xc = 0

- Add the first equation to the second. Subtract twice the first from the third, and three times the first from the fourth.
- Then subtract five times the new second equation from the new third, and four times it from the new fourth.

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ -1 & 3 & 0 & 0 \\ 2 & 1 & 7 & 0 \\ 3 & -2 & 7 & 0 \end{array}\right] \Rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 5 & 5 & 0 \\ 0 & 4 & 4 & 0 \end{array}\right] \Rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

- The row echelon form involves the free variable c_3.
- Therefore, the homogeneous system has nontrivial solutions, and x_1, x_2, x_3 are linearly dependent.

Example 6 Interpreting the Free Variable

- The nonzero equations in the row echelon form are $c_1 - 2c_2 + c_3 = 0$ and $c_2 + c_3 = 0$.
- The second equation gives $c_2 = -c_3$.
- Substituting in the first gives $c_1 = -3c_3$.
- Hence c_3 may be chosen freely, with the other coefficients determined by $c_1 = -3c_3$ and $c_2 = -c_3$.
- For $c_3 = 1$, we obtain the nontrivial solution $c_1 = -3$, $c_2 = -1$, $c_3 = 1$, giving $-3x_1 - x_2 + x_3 = 0$.
- This verifies the dependency conclusion directly.
- We now ask what independence implies about representation.

Theorem 3.3.2 Unique Linear Combinations

Let $v_1, \ldots, v_n$ be vectors in a vector space V. A vector v in Span($v_1, \ldots, v_n$) can be written uniquely as a linear combination of $v_1, \ldots, v_n$ if and only if $v_1, \ldots, v_n$ are linearly independent.

- Membership in the span guarantees that a representation exists.
- Independence guarantees that its coefficients are unique.
- If the vectors are dependent, a linear combination with coefficients not all zero equals zero. Adding it changes the coefficients without changing v.

Theorem 3.3.2 Setting Up the Proof

If $v \in \text{Span}(v_1, \ldots, v_n)$, then $v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$. (5)

Suppose that v can also be expressed as $v = \beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_n v_n$. (6)

We will show that, if $v_1, \ldots, v_n$ are linearly independent, then $\beta_i = \alpha_i$ for $i = 1, \ldots, n$.

If $v_1, \ldots, v_n$ are linearly dependent, it is possible to choose the β_i's different from the α_i's.

Both equations represent the same vector v; the question is whether their coefficients must agree.

Theorem 3.3.2 Independence Gives Uniqueness

Assume that $v_1, \ldots, v_n$ are linearly independent.

Subtracting (6) from (5) yields $(\alpha_1 - \beta_1)v_1 + (\alpha_2 - \beta_2)v_2 + \cdots + (\alpha_n - \beta_n)v_n = 0$. (7)

By linear independence, all coefficients in (7) must be zero:

$$\alpha_1 - \beta_1 = 0, \alpha_2 - \beta_2 = 0, \ldots, \alpha_n - \beta_n = 0.$$

Hence, $\alpha_1 = \beta_1, \alpha_2 = \beta_2, \ldots, \alpha_n = \beta_n$.

Thus, representation (5) is unique when $v_1, \ldots, v_n$ are linearly independent.

Any two representations have identical coefficients; therefore, no different representation is possible.

Theorem 3.3.2 Dependence Changes the Coefficients

If $v_1, \ldots, v_n$ are linearly dependent, there exist $c_1, \ldots, c_n$, not all zero, such that

$$0 = c_1v_1 + c_2v_2 + \cdots + c_nv_n. \quad (8)$$

Set $\beta_1 = \alpha_1 + c_1, \beta_2 = \alpha_2 + c_2, \ldots, \beta_n = \alpha_n + c_n$.

Adding (5) and (8) gives

$$\begin{aligned} v &= (\alpha_1 + c_1)v_1 + (\alpha_2 + c_2)v_2 + \cdots + (\alpha_n + c_n)v_n \\ &= \beta_1v_1 + \beta_2v_2 + \cdots + \beta_nv_n. \end{aligned}$$

Adding a combination equal to zero leaves v unchanged, while the coefficients may change.

Theorem 3.3.2 Completing the Proof

Since the c_i's are not all zero, $\beta_i \neq \alpha_i$ for at least one i.

We have obtained two different coefficient lists that represent the same vector v.

Thus, if $v_1, \ldots, v_n$ are linearly dependent, the representation of v as a linear combination of $v_1, \ldots, v_n$ is not unique.

Together with the independent case, this proves the theorem.

The theorem applies in any vector space V.

How Do We Test Independence of Polynomials and Functions?

Linear Independence in P_n

- P_n is the vector space of all polynomials of degree less than n.
- To test whether $p_1, p_2, \ldots, p_k$ are linearly independent in P_n, set

$$c_1p_1 + c_2p_2 + \cdots + c_kp_k = z. \quad (9)$$

- Here z represents the zero polynomial:

$$z(x) = 0x^{n-1} + 0x^{n-2} + \cdots + 0x + 0.$$

- Rewrite the polynomial on the left-hand side as

$$a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n.$$

Equating Polynomial Coefficients

- Two polynomials are equal if and only if their coefficients are equal.
- Thus, equality with the zero polynomial requires $a_1 = a_2 = \cdots = a_n = 0$.
- Each a_i is a linear combination of $c_1, c_2, \ldots, c_k$.
- These conditions give a homogeneous linear system with unknowns $c_1, c_2, \ldots, c_k$.
- If the system has only the trivial solution, the polynomials are linearly independent; otherwise, they are linearly dependent.
- This is the same test used for vectors in $\mathbb{R}^n$.

Example 7 Polynomial Independence

To test whether the following vectors are linearly independent, let

$$p_1(x) = x^2 - 2x + 3, \quad p_2(x) = 2x^2 + x + 8, p_3(x) = x^2 + 8x + 7.$$

- Set $c_1p_1(x) + c_2p_2(x) + c_3p_3(x) = 0x^2 + 0x + 0$.
- Expanding and grouping terms by powers of x gives

  $(c_1 + 2c_2 + c_3)x^2 + (-2c_1 + c_2 + 8c_3)x + (3c_1 + 8c_2 + 7c_3) = 0x^2 + 0x + 0$.
- The coefficients of x^2, x, and the constant term must each be zero.

Example 7 The Homogeneous System

- Equating coefficients leads to the system

$$\begin{aligned} c_1 + 2c_2 + c_3 &= 0 \\ -2c_1 + c_2 + 8c_3 &= 0 \\ 3c_1 + 8c_2 + 7c_3 &= 0. \end{aligned}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ -2 & 1 & 8 \\ 3 & 8 & 7 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Expanding the first row gives $1(7-64)-2(-14-24)+(-16-3)=0$.

The coefficient matrix is singular, so there are nontrivial solutions.

Therefore p_1, p_2, and p_3 are linearly dependent.

Example 7 A Nontrivial Solution

- The first equation gives $c_1 = -2c_2 - c_3$.
- Substitution into the other two equations gives $5c_2 + 10c_3 = 0$ and $2c_2 + 4c_3 = 0$.
- Hence $c_2 = -2c_3$ and $c_1 = 3c_3$.
- Choosing $c_3 = 1$ gives $c_1 = 3$ and $c_2 = -2$.
- These coefficients are not all zero, and

  $3p_1(x) - 2p_2(x) + p_3(x)$

  $= (3 - 4 + 1)x^2 + (-6 - 2 + 8)x + (9 - 16 + 7)$

  $= 0x^2 + 0x + 0.$
- This explicitly verifies the dependence concluded from the determinant.

Linear Dependence in $C^{(n-1)}[a, b]$

- Determinants can also help decide whether n functions are linearly independent in $C^{(n-1)}[a, b]$.
- Let $f_1, f_2, \ldots, f_n$ be elements of $C^{(n-1)}[a, b]$.
- If they are linearly dependent, there exist scalars $c_1, c_2, \ldots, c_n$, not all zero, such that $c_1f_1(x) + c_2f_2(x) + \cdots + c_nf_n(x) = 0$ (10) for each x in [a, b]. The c_i are constants, independent of x.
- Taking the derivative of both sides with respect to x yields $c_1f_1'(x) + c_2f_2'(x) + \cdots + c_nf_n'(x) = 0$.

The System Obtained by Differentiation

- Continue taking derivatives of both sides of (10).
- Because the functions are in $C^{(n-1)}[a, b]$, these derivatives exist.
- We obtain the homogeneous system

$$
\begin{aligned}
&c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0 \\
&c_1 f_1'(x) + c_2 f_2'(x) + \cdots + c_n f_n'(x) = 0 \\
&\vdots \\
&c_1 f_1^{(n-1)}(x) + c_2 f_2^{(n-1)}(x) + \cdots + c_n f_n^{(n-1)}(x) = 0
\end{aligned}
$$

- The same constants $c_1, c_2, \ldots, c_n$, not all zero, satisfy every equation for every x in [a, b].

The Matrix Equation for Each Fixed x

- For each fixed x in [a, b], equation (11) is

$$\begin{bmatrix} f_1(x) & f_2(x) & \cdots & f_n(x) \\ f_1'(x) & f_2'(x) & \cdots & f_n'(x) \\ \vdots & & & \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x) \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

- It has the same nontrivial solution $(c_1, c_2, \ldots, c_n)^T$: $\alpha_1 = c_1, \alpha_2 = c_2, \ldots, \alpha_n = c_n$.
- Thus, if the functions are linearly dependent, the coefficient matrix is singular for each fixed x in [a, b].

Dependence Forces a Zero Determinant

- A singular matrix has determinant zero.
- Therefore, if f_1, f_2, …, f_n are linearly dependent,

$$\begin{vmatrix} f_1(x) & f_2(x) & \cdots & f_n(x) \\ f_1'(x) & f_2'(x) & \cdots & f_n'(x) \\ \vdots & & & \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x) \end{vmatrix} = 0$$

  for every x in [a, b].
- The determinant depends on x, so it defines a function of x.
- The text gives this determinant function a name in the next definition.

Definition of the Wronskian

Let $f_1, f_2, \ldots, f_n$ be functions in $C^{(n-1)}[a, b]$, and define the function $W[f_1, f_2, \ldots, f_n](x)$ in $[a, b]$ by

$$W[f_1, f_2, \cdots, f_n](x) = \begin{vmatrix} f_1(x) & f_2(x) & \cdots & f_n(x) \\ f_1'(x) & f_2'(x) & \cdots & f_n'(x) \\ \vdots & & & \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x) \end{vmatrix}$$

The function $W[f_1, f_2, \ldots, f_n](x)$ is called the **Wronskian** of $f_1, f_2, \ldots, f_n$.

Each column comes from one function; successive rows contain its successive derivatives.

Theorem 3.3.3 A Wronskian Test

Let $f_1, f_2, \ldots, f_n$ be elements of $C^{(n-1)}[a, b]$. If there exists a points x_0 in [a, b] such that $W[f_1, f_2, \ldots, f_n](x_0) \neq 0$, then $f_1, f_2, \ldots, f_n$ are linearly independent.

Proof

If $f_1, f_2, \ldots, f_n$ were linearly dependent, then by the preceding discussion, the coefficient matrix in (11) would be singular for each x in $[a, b]$ and hence $W[f_1, f_2, \ldots, f_n](x)$ would be identically zero on $[a, b]$.

$$\begin{bmatrix} f_1(x) & f_2(x) & \cdots & f_n(x) \\ f_1'(x) & f_2'(x) & \cdots & f_n'(x) \\ \vdots & & & \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x) \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Example 8 Exponential Functions

Show that e^x and e^{-x} are linearly independent in $C(-\infty, \infty)$.

$$W\left[e^x, e^{-x}\right] = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^x \end{vmatrix} = -2$$

Indeed, $e^x(-e^{-x}) - e^{-x}e^x = -1 - 1 = -2$.

The Wronskian is not identically zero; it is nonzero at every x.

Theorem 3.3.3 gives linear independence on every finite interval.

Any relation valid on all real x would also hold on such an interval;

hence the functions are linearly independent in $C(-\infty, \infty)$.

Example 9 The Required Derivatives

Consider the functions x^2 and $x|x|$ in $C[-1, 1]$.

Both belong to $C^1[-1, 1]$, so we can compute their Wronskian.

The derivative of x^2 is $2x$, a continuous function.

For $x > 0$, $x|x| = x^2$, so its derivative is $2x = 2|x|$.

For $x < 0$, $x|x| = -x^2$, so its derivative is $-2x = 2|x|$.

At $x = 0$, the derivative is $\lim_{x \to 0} \frac{x|x| - 0}{x - 0} = \lim_{x \to 0} |x| = 0$.

Thus, the derivative of $x|x|$ is $2|x|$ throughout $[-1, 1]$.

This derivative is continuous, including at $x = 0$.

Example 9 An Identically Zero Wronskian

Using the derivatives just established, we obtain

$$W[x^2, x|x|] = \begin{vmatrix} x^2 & x|x| \\ 2x & 2|x| \end{vmatrix} \equiv 0$$

Indeed, $x^2(2|x|) - (x|x|)(2x) = 0$ for every x in $[-1, 1]$.

The symbol $\equiv 0$ means identically zero on the interval.

Since the Wronskian is identically zero, it gives no information as to whether the functions are linearly independent.

We must return to the definition of linear independence.

Example 9 Testing the Functions Directly

Suppose $c_1x^2 + c_2x|x| = 0$ for all x in $[-1, 1]$.

In particular, evaluating at $x = 1$ and $x = -1$ gives

$$c_1 + c_2 = 0$$

$$c_1 - c_2 = 0.$$

Adding the equations gives $2c_1 = 0$, so $c_1 = 0$.

Substitution then gives $c_2 = 0$.

Thus, x^2 and $x|x|$ are linearly independent in $C[-1, 1]$, even though $W[x^2, x|x|] \equiv 0$.

This example shows that the converse of Theorem 3.3.3 is not valid.

Example 10 Polynomial Functions

Show that the vectors 1, x, x^2, and x^3 are linearly independent in $C((-\infty, \infty))$.

$$W[1, x, x^2, x^3] = \begin{vmatrix} 1 & x & x^2 & x^3 \\ 0 & 1 & 2x & 3x^2 \\ 0 & 0 & 2 & 6x \\ 0 & 0 & 0 & 6 \end{vmatrix} = 12$$

The matrix is upper triangular, so its determinant is

$1 \cdot 1 \cdot 2 \cdot 6 = 12$. The Wronskian is not identically zero.

Thus, the vectors are linearly independent in $C(-\infty, \infty)$.

How Do Bases and Dimension Describe a Vector Space?

Basis and Dimension

Definition

The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ form a **basis** for a vector space V if and only if

(1) $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent (minimal spanning set)

(2) $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ span V (spanning set)

Both conditions are required: independence makes the spanning set minimal, and spanning reaches every vector in V.

Example 1 Bases for $\mathbb{R}^3$

- The *standard basis* for R^3 is $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$; however, there are many bases that we could choose for R^3.
- For example,

$$\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}, \begin{bmatrix}2\\0\\1\end{bmatrix}\right\} \text{ and } \left\{\begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}\right\}$$

are both bases for R^3. We will see shortly that any basis for $\mathbb{R}^3$ must have exactly three elements.

Example 1 Verifying Both Bases

- Use the given vectors as the columns of a square matrix.
- For the first set and the second set, respectively,

$$\begin{vmatrix} 1 & 0 & 2 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = 1, \qquad \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = -1.$$

- Both determinants are nonzero. The corresponding homogeneous systems therefore have only the trivial solution, so each set of columns is linearly independent.
- Each matrix is also nonsingular. Every vector in $\mathbb{R}^3$ can therefore, be expressed as a linear combination of its columns.
- Each set spans $\mathbb{R}^3$ and satisfies both conditions for a basis.

Example 2 A Basis for $\mathbb{R}^{2\times2}$

In $\mathbb{R}^{2\times2}$, consider the set $\{E_{11}, E_{12}, E_{21}, E_{22}\}$, where

$$E_{11}=\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}, \quad E_{12}=\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$$

$$E_{21}=\begin{bmatrix}0 & 0\\ 1 & 0\end{bmatrix}, \quad E_{22}=\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}$$

Each matrix has one entry equal to 1 and all other entriesequal to 0.

Multiplying it by a scalar specifies that entry.

We verify independence and spanning using matrix entries.

Example 2 Independence and Spanning

- If $c_1E_{11} + c_2E_{12} + c_3E_{21} + c_4E_{22} = O$, then $\begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

- Equality of corresponding entries gives $c_1 = c_2 = c_3 = c_4 = 0$.
- Therefore, E_{11}, E_{12}, E_{21}, and E_{22} are linearly independent.
- If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is in $R^{2\times 2}$, then

$$A = aE_{11} + bE_{12} + cE_{21} + dE_{22}$$

- Thus, E_{11}, E_{12}, E_{21}, E_{22} span $\mathbb{R}^{2\times 2}$, and hence form a basis for $\mathbb{R}^{2\times 2}$.

A Basis for a Null Space

- A basis can also describe a particular subspace.

  To find all solutions of the homogeneous system

  $x_1 + x_2 + x_3 = 0$

  $2x_1 + x_2 + x_4 = 0$

- As in Example 9 of Section 3.2, the equations give

  $x_2 = -2x_3 + x_4$ and $x_1 = x_3 - x_4$.

- The variables x_3 and x_4 are free; x_1 and x_2 depend on them.

- Set $x_3 = \alpha$ and $x_4 = \beta$.

The Solution in Example 9 and Its Span

- The solution in Example 9 of Section 3.2 is

$$x = \begin{bmatrix} \alpha - \beta \\ -2\alpha + \beta \\ \alpha \\ \beta \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

- The coefficients α and β are arbitrary real numbers.
- Therefore, these two vectors span N(A), as stated in Example 9 of Section 3.2.
- We next verify their linear independence.

The Null Space Spanning Set Is a Basis

- To test the two spanning vectors for independence, set

$$c_1 \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix} = 0.$$

- The third entry of this equation gives $c_1 = 0$.
- The fourth entry gives $c_2 = 0$.
- Thus, the two vectors are linearly independent.
- They span N(A) and form a basis for N(A).
- A basis describes every solution of the homogeneous system.

Theorem 3.4.1

If $\{v_1, v_2, \ldots, v_n\}$ is a spanning set for a vector space V, then any collection of m vectors in V, where m > n, is linearly dependent.

Proof

Let $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m$ be m vectors in V, when $m > n$. Then, since $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ span V, we have $\mathbf{u}_i = a_{i1}\mathbf{v}_1 + a_{i2}\mathbf{v}_2 + \cdots a_{in}\mathbf{v}_n \quad \text{for} \quad i = 1, 2, ..., m$

Each $\mathbf{u}_i$ has a representation using the same n spanning vectors.

We will choose coefficients $c_1, \ldots, c_m$, not all zero, so that

$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_m\mathbf{u}_m = 0.$

Proof of Theorem 3.4.1 Regrouping Terms

Substitute the representation of each u_i into the combination:

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_m\mathbf{u}_m = c_1 \sum_{j=1}^{n} a_{1j}\mathbf{v}_j + c_2 \sum_{j=1}^{n} a_{2j}\mathbf{v}_j + \cdots + c_m \sum_{j=1}^{n} a_{mj}\mathbf{v}_j$$

Rearranging terms, we obtain

$$\sum_{i=1}^{m}\left[c_i\left(\sum_{j=1}^{n} a_{ij}\mathbf{v}_j\right)\right] = \sum_{j=1}^{n}\left(\sum_{i=1}^{m} a_{ij}c_i\right)\mathbf{v}_j$$

The inner sum on the right is the coefficient of v_j.

Set each coefficient equal to zero to obtain n equations in $c_1, \ldots, c_m$.

Proof of Theorem 3.4.1 A Nontrivial Solution

Now consider the system of equations $\sum_{i=1}^{m} a_{ij}c_i = 0, \quad j = 1, 2, \ldots, n.$

There are m unknowns and n equations. Because m > n, Theorem 1.2.1 gives a nontrivial solution $(\hat{c}_1, \hat{c}_2, \ldots, \hat{c}_m)^T$.

For this solution, every coefficient of v_j is zero, so

$$\hat{c}_1\mathbf{u}_1 + \hat{c}_2\mathbf{u}_2 + \cdots + \hat{c}_m\mathbf{u}_m = \sum_{j=1}^{n} 0\mathbf{v}_j = \mathbf{0}.$$

At least one $\hat{c}_i$ is nonzero; hence, $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m$ are dependent.

The spanning set need not itself be independent: n spanning vectors already limit an independent set to at most n vectors.

Corollary 3.4.2

If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ are both bases for a vector space V, then $n = m$.

Proof

Let both $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ be abses for V. Since $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ span V and $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m$ are linearly independent, it follows from Theorem 3.4.1 that $m \leq n$. By same reasoning, $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m$ span V and $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent, so $n \leq m$.

Dimension

Definition

Let V be vector space. If V has a basis consisting of n vectors, we say that V has **dimension** n. The subspace $\{\mathbf{0}\}$ of V is said to have dimension 0. V is said to be **finite-dimensional** if there is a finite set of vectors that spans V; otherwise we say that V is **infinite-dimensional**.

- The notation dim V gives the number of vectors in any basis.
- The two bases in Example 1 each contain three vectors, so dim $\mathbb{R}^3 = 3$. Example 2 gives dim $\mathbb{R}^{2\times2} = 4$.
- The basis for N(A) above has two vectors, so dim N(A) = 2.
- Dimension counts basis elements, including when the elements are matrices rather than column vectors.

Dimension in $\mathbb{R}^3$

(a) Span(x) = {αx | α is a scalar} is a line if x ≠ 0.

(b) Span(x, y) = {αx + βy | α and β are scalars} is a plane if x and y are linearly independent.

(c) Span(x, y, z) = $\mathbb{R}^3$

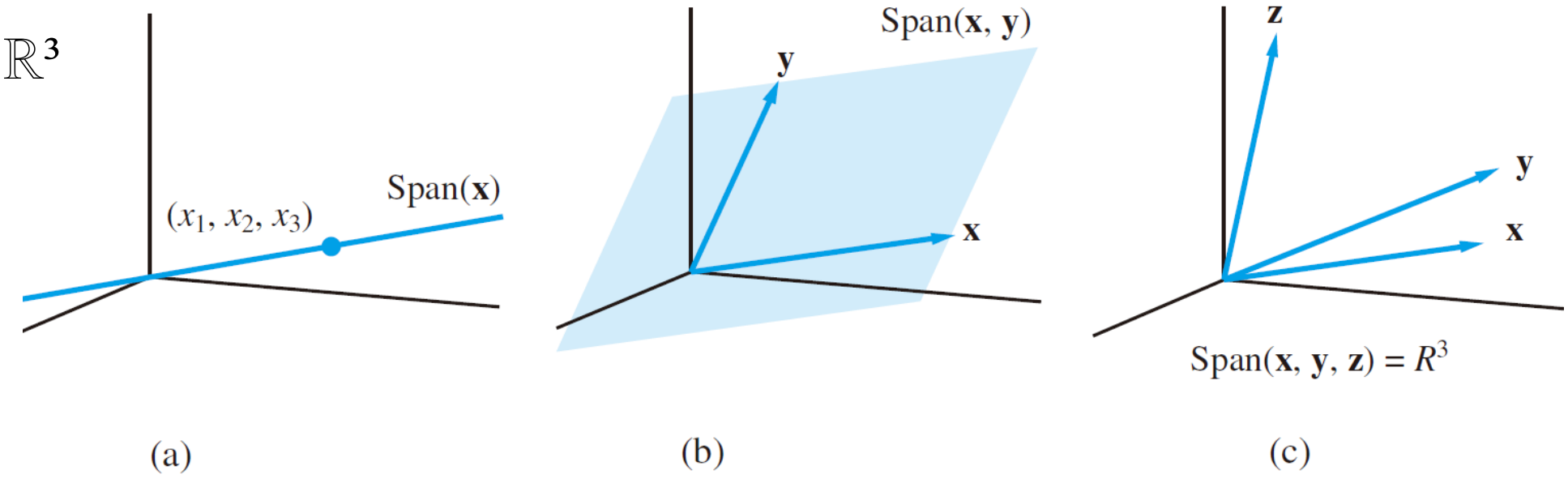


Figure 3.4.1.

Example 3 An Infinite Dimensional Space

- Let P be the vector space of all polynomial. We claim that P is infinite dimensional. If P were finite dimensional, say, if dimension n, any set of n+1 vectors would be linearly dependent.
- However, $1, x, x^2, \ldots, x^n$ are linearly independent, since $W[1, x, x^2, \ldots, x^n] > 0$. Therefore, P cannot be of dimension n. Since m was arbitrary, P must be infinite dimension. The same argument shows that $C[a, b]$ is infinite dimensional.

Theorem 3.4.3

If V is a vector space of dimension $n > 0$, then

(I) any set of n linearly independent vectors spans V.

(II) any n vectors that span V are linearly independent.

- A basis requires both independence and spanning. Once dim V = n is known and a set has exactly n vectors, this theorem shows that verifying either condition is enough.
- Part (I) supplies spanning from independence; part (II) supplies independence from spanning.
- We prove each statement using the limit on independent sets established in Theorem 3.4.1.

Proof of Theorem 3.4.3 Part I

Suppose $v_1, \ldots, v_n$ are linearly independent and v is any other vector in V. Since dim V = n, a basis has n vectors

that span V. Theorem 3.4.1 makes $v_1, \ldots, v_n$, v dependent.

Thus, scalars $c_1, \ldots, c_n, c_{n+1}$, not all zero, satisfy $c_1v_1 + c_2v_2 + \cdots + c_nv_n + c_{n+1}v = 0$. (1)

The scalar c_{n+1} cannot be zero. Otherwise, (1) would give a nontrivial dependence among $v_1, \ldots, v_n$. Hence, $v = \alpha_1v_1 + \alpha_2v_2 + \cdots + \alpha_nv_n$, where $\alpha_i = -c_i/c_{n+1}$ for $i = 1, 2, \ldots, n$.

Since v was arbitrary, $v_1, \ldots, v_n$ span V.

Together with their independence, this makes them a basis.

Proof of Theorem 3.4.3 Part II

Suppose $v_1, \ldots, v_n$ span V. If they were linearly dependent, one vector, say v_n, could be written as a linear combination of the others. Then $v_1, \ldots, v_{n-1}$ would still span V.

If the remaining vectors were dependent, another vector could be eliminated while preserving spanning. Continue until a linearly independent spanning set remains.

This set would be a basis with $k < n$ elements.

But Corollary 3.4.2 says every basis has the same number of elements, contradicting dim V = n.

Therefore, $v_1, \ldots, v_n$ must be linearly independent.

The n spanning vectors are consequently a basis for V.

Example 4 Using Dimension to Verify a Basis

Show that $\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}-2\\1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}\right\}$ is a basis for $\mathbb{R}^3$

Since dim $\mathbb{R}^3 = 3$, we need only show that these three vectors are linearly independent.

$$\begin{vmatrix}1 & -2 & 1\\2 & 1 & 0\\3 & 0 & 1\end{vmatrix} = 1 + 4 - 3 = 2 \neq 0.$$

The determinant is nonzero, so the vectors are linearly independent. By Theorem 3.4.3(I), the three vectors also span $\mathbb{R}^3$. They therefore form a basis for $\mathbb{R}^3$.

Theorem 3.4.4

If V is a vector space of dimension $n > 0$, then

(i) no set of fewer than n vectors can span V.

(ii) any subset of fewer than n linearly independent vectors can be extended to form a basis for V.

(iii) any spanning set containing more than n vectors can be pared down to form a basis for V.

Proof of Theorem 3.4.4 Part i

A basis for V consists of n linearly independent vectors.

Suppose a set of k vectors, where $k < n$, spanned V.

Theorem 3.4.1 would then make every collection of more than k vectors in V linearly dependent.

In particular, the n basis vectors would be dependent, because $n > k$. This contradicts the definition of a basis.

Therefore, no set of fewer than n vectors can span V.

This is the same spanning-set argument used in

Theorem 3.4.3(I). It explains why an independent set with fewer than n vectors still needs additional vectors to span V.

Proof of Theorem 3.4.4 Part ii

Suppose $v_1, \ldots, v_k$ are linearly independent and $k < n$.

By part (i), Span($v_1, \ldots, v_k$) is a proper subspace of V.

Choose v_{k+1} in V but not in Span($v_1, \ldots, v_k$).

The vectors $v_1, \ldots, v_k, v_{k+1}$ are linearly independent: if they were dependent, v_{k+1} would be a linear combination

of the independent vectors $v_1, \ldots, v_k$, contrary to its choice.

If $k + 1 < n$, choose another vector outside their span.

Continue until $\{v_1, \ldots, v_k, v_{k+1}, \ldots, v_n\}$ has n linearly independent vectors.

Theorem 3.4.3(I) says they span V.

Thus, the original independent set has been extended to form a basis for V.

Proof of Theorem 3.4.4 Part iii

Suppose $v_1, \ldots, v_m$ span V and $m > n$.

By Theorem 3.4.1, these vectors are linearly dependent.

One vector, say v_m, is a linear combination of the others.

Eliminate v_m. The remaining $m - 1$ vectors still span V, because every occurrence of v_m can be replaced by that linear combination.

If $m - 1 > n$, eliminate another dependent vector.

Continue until a spanning set with n vectors remains.

By Theorem 3.4.3(II), these n vectors are linearly independent.

The remaining set is therefore a basis for V.

This gives a basis by paring down the original spanning set.

Standard Bases

- The standard basis for $\mathbb{R}^n$ is $\{e_1, e_2, \ldots, e_n\}$.
- The standard basis for $\mathbb{R}^{2\times 2}$ is $\{E_{11}, E_{12}, E_{21}, E_{22}\}$.
- The standard basis for P_n is $\{1, x, x^2, x^3, \ldots, x^{n-1}\}$.

- Recall that P_n consists of polynomials of degree less than n, together with the zero polynomial. Its standard basis has n elements, so dim P_n = n. This differs from P, the space of all polynomials, which is infinite dimensional.
- Standard bases are the most natural ones for representing vectors, matrices, and polynomials. Other bases may be more appropriate for applied problems.

Thank You!